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To the Editors, *PRX Quantum*

Re: Submission of "Scrambling, not athermality, protects quantum information under generic measurement: chaotic states outperform quantum many-body scar codes"

On July 10, 2026, we submit for your consideration the manuscript above (research article, two authors) together with Supplemental Material (one PDF: eight sections, three figures, one table) and an openly available simulation package reproducing every result.

A widely shared intuition holds that quantum many-body scars—non-thermal, weakly entangled, unusually long-lived—should make especially good quantum memories. We show that this intuition is wrong, and in doing so identify what actually makes a many-body code robust to measurement. Computing the reference-qubit coherent information of scar versus thermal codes in a monitored circuit, we find that generic chaotic states protect information *better* under every realistic measurement and at every accessible size in the paradigmatic PXP model—down to the largest code an entire scar subspace can hold—while an exactly solvable second model confirms the mechanism and shows that scars gain an advantage only under a single fine-tuned probe. The result is counterintuitive by construction: the very structure that singles scars out is precisely what a measurement exploits to read them out.

Two features give the work reach well beyond scars. First, it yields a general design principle for measurement-robust quantum memory: protection is set by a code's spread in the measurement basis, not by any thermodynamic property, and structured subspaces whose symmetry generators couple the encoded states are actively harmful. We make this falsifiable—an exactly solvable model isolates a single control parameter, a Casimir variance, and predicts correctly the unique fine-tuned measurement under which scars can be rescued. Second, of immediate practical value to the many groups now benchmarking codes and channels, we document a reproducibility trap: comparing against a single natural-looking reference eigenstate inverted our own initial conclusion, and only ensemble averaging exposed it as an outlier—a cautionary result we consider broadly useful in its own right. Together these carry the work past any mere confirmation of the thermalization–error-correction correspondence: they turn that correspondence into an operational, testable statement about code *design*, overturn the opposing folklore with a decisive exact-model test, and flag a pitfall in how such statements are established.

In brief, relative to prior work: *(i)* the first head-to-head coding comparison of scar versus thermal subspaces under monitoring, at the level of channel capacity rather than entanglement or fidelity; *(ii)* a two-channel protection criterion with an exact single-shot predictor, validated across 65 code–measurement pairs in two models; *(iii)* a documented sign-reversing baseline artifact of direct relevance to anyone benchmarking codes against individual eigenstates.

We believe *PRX Quantum* is the right venue. The result speaks to the whole quantum-information community working on monitored dynamics, error correction, and quantum memory; it is conceptual and general rather than model-specific; and it is delivered with full reproducibility—an open package regenerates every figure with one command, and all quantitative claims have been independently checked against primary sources. We hope the Editors will agree that overturning a common intuition, extracting a design principle, and surfacing a shared methodological hazard together meet the journal's bar for significance and breadth.

We suggest the following referees, all leading experts in measurement-induced dynamics, quantum error correction, or quantum many-body scars (institutional contact addresses are publicly listed):

- Prof. Matthew P. A. Fisher, University of California, Santa Barbara — measurement-induced transitions
- Prof. Ehud Altman, University of California, Berkeley — error correction and monitored circuits
- Dr. Michael J. Gullans, NIST / University of Maryland — purification transitions
- Prof. Vedika Khemani, Stanford University — random and monitored circuits
- Prof. Adam Nahum, École Normale Supérieure, Paris — entanglement dynamics
- Prof. Zlatko Papić, University of Leeds — quantum many-body scars
- Prof. Maksym Serbyn, IST Austria — quantum many-body scars
- Dr. Sanjay Moudgalya, Technical University of Munich — exact scars and spectrum-generating algebras

We do not request the exclusion of any referee.

On behalf of both authors,
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